André F. Rendeiro

Principal Investigator, Computational Biologist

arendeiro@cemm.oeaw.ac.at
andre-rendeiro.com
ORCID: 0000-0002-7635-3473
afrendeiro
afrendeiro

Current positions

2022/06 - **Principal Investigator**, *Research Center for Molecular Medicine of the Austrian Academy of Sciences*, Austria

2024/03 - **Research Group Leader**, *Boltzmann Institute for Network Medicine at the University of Vienna*, Austria

Past positions and education

2020/03 - 2022/04 **Postdoctoral Associate in Computational Biomedicine**, *Institute of Computational Biomedicine, Englander Institute for Precision Medicine, Weill Cornell Medicine*, USA
Supervisor: Olivier Elemento

2014/09 - 2020/01 **PhD**, *Medical University of Vienna*, Austria
Supervisor: Christoph Bock

2012/09 - 2014/06 **MSc in Molecular and Cell Biology**, *University of Aveiro*, Portugal

2008/09 - 2012/07 **BSc in Biology**, *University of Aveiro*, Portugal

Key research

1. Zheng et al. Ω **LazySlide: accessible and interoperable whole slide image analysis**. Nature Methods (2026). doi:10.1038/s41592-026-03044-7
2. Buljan et al. Ω **Systematic mapping of human tissue microanatomy reveals age-associated remodeling and resilience**. BioRxiv (2025). doi:10.1101/2025.10.13.682110
3. Abila, Buljan, Zheng et al. Ω **Tissue clocks derived from histological signatures of biological aging enable tissue-specific aging predictions from blood**. BioRxiv (2024). doi:10.1101/2024.11.14.618081
4. Kim et al. Ω **Unsupervised discovery of tissue architecture in multiplexed imaging**. Nature Methods (2022). doi:10.1038/s41592-022-01657-2
5. Rendeiro*, Ravichandran* et al. **The spatial landscape of lung pathology during COVID-19 progression**. Nature (2021). doi:10.1038/s41586-021-03475-6

Ω *corresponding author*; * *first-author*

Grants, fellowships and awards

2026 **'Beyond the Journal' award**, *Experiment Foundation*

2026 **Young Academy member**, *Austrian Academy of Sciences (OeAW)*, Member

2025 - 2030 **Starting Grant**, *European Research Council (ERC)*, co-PI

2024 - 2028 **Understanding Biology with AI/ML 2023 Life Sciences call**, *Vienna Science and Technology Fund (WWTF)*, co-PI

2024 - 2031 **Ludwig Boltzmann Institute for Network Medicine**, *Ludwig Boltzmann Association (LBG)*, co-PI

2022 - 2027 **Aging and Longevity research**, *Angelini Ventures S.p.A. Rome, Italy*, co-PI

2020 - 2022 **Molecular and Translational Oncology Research (MTOR) training program**, *National Cancer Institute, USA*, Postdoc

2013 - 2014 **Erasmus studies mobility program scholarship**, *European Commission*, Master studies

2011 - 2012 **Erasmus studies mobility program scholarship**, *European Commission*, Undergrad studies

2009 - 2010 **'Integration into Research' Grant**, *Science and Technology Foundation, Portugal*, Undergrad studies

Publications

Publications: 57 (45 peer reviewed, 9 preprints, 12 first-author, 3 last-author)

Citations: 8708 (last 5 years: 7360)

h-index: 34 (last 5 years: 33)

Google Scholar Profile: https://scholar.google.com/citations?user=lj17pqEAAAAJ

* *equal first-author contributions* Ω *joint corresponding authors*

Preprints (does not include preprints later published in peer-reviewed journals)

9. Daphne A Superville, Eva Chrenková, Yimin Zheng, Alexis J. Combes, Iros Barozzi, Zena Werb, André F. Rendeiro, Christopher S. McGinnis, Andrei Goga, Juliane Winkler. **Tumor Heterogeneity Induces Pro- and Anti-metastatic Myeloid Cell Phenotypes in Breast Cancer Lung Metastasis**. BioRxiv (2026).doi:10.64898/2026
8. Iva Buljan, Zsuzsanna Bagó-Horváth, André F. Rendeiro. **Systematic mapping of human tissue microanatomy reveals age-associated remodeling and resilience**. BioRxiv (2025).doi:10.1101/2025.10.13.682110
7. Sarah Dobner, Lisa Kleissl, Fanni Tóth, Robert Paxton, Irena Yordanova, Cedric Vanluyten, Hanna Toth, Aglaja Kopf, Jake Burton, Dunja Al-Nuaimi, Vera Belyaeva, André F. Rendeiro, Christoph Bock, Selma Osmanagic-Myers, Joanna Kalucka, Laurens Ceulemans4, Georg Stary, Abdel Rahman Abdel Fattah, Laura P.M.H. de Rooij. **Uncovering the transcriptional hallmarks of endothelial cell aging via integrated single-cell analysis**. BioRxiv (2025).doi:10.1101/2025.08.18.669055
6. Jiwoon Park, Roberto De Gregorio, Erika Hissong, Elif Ozcelik, Nicholas Bartelo, Felipe Segato Dezem, Luke Zhang, Maycon Marção, Hannah DuBose, Yimin Zheng, Ernesto Abila, Junbum Kim, Jacqueline Proszynski, Akua A Agyemang, Mohith Reddy Arikatla, Evelyn Metzger, Stefan Rogers, Prajan Divakar, Parambir S Dulai, Jason W Reeves, Yan Liang, Liuliu Pan, Sayani Bhattacharjee, Kimberly Young, Ashley Heck, Mithra Korukonda, Dan McGuire, Lidan Wu, Aster Wardhani, Joseph Beechem, George Church, Steven M Lipkin, Sanjay Patel, Fabio Socciarelli, Sebastien Monette, Brian Robinson, Massimo Loda, Olivier Elemento, Luciano Martelotto, Jasmine T Plummer, André F. Rendeiro, Alicia Alonso, Robert E Schwartz, Shauna L Houlihan, Christopher E Mason. **The Spatial Atlas of Human Anatomy (SAHA): A Multimodal Subcellular-Resolution Reference Across Human Organs**. BioRxiv (2025).doi:10.1101/2025.06.16.658716
5. Adrien Grimont, David J Falvo, Whitney J Sisso, Paul Zumbo, Christopher W Chan, Francisco Santos, Grace Pan, Megan Cleveland, Tomer Yaron, Alexa S Osterhoudt, Yinuo Meng, Maria Paz Zafra, William B Fall, André F. Rendeiro, Erika Hissong, Rhonda K Yantiss, Doron Betel, Mark A Magnuson, Steven D Leach, Anil K Rustgi, Lukas E Dow, Rohit Chandwani. **Molecular dynamics driving phenotypic divergence among KRAS mutants in pancreatic tumorigenesis**. BioRxiv (2025).doi:10.1101/2025.05.28.656689
4. Salvo Danilo Lombardo, André F. Rendeiro, Jörg Menche. **A multilayer network approach elucidates time- and tissue-specific developmental and aging processes**. BioRxiv (2025).doi:10.1101/2025.05.02.651926
3. Ernesto Abila, Iva Buljan, Yimin Zheng, Tamas Veres, Zhilong Weng, Maja C Nackenhorst, Wolfgang Hulla, Yuri Tolkach, Adelheid Woehrer, André F. Rendeiro$^{\Omega}$. **Tissue clocks derived from histological signatures of biological aging enable tissue-specific aging predictions from blood**. BioRxiv (2024).doi:10.1101/2024.11.14.618081
2. Everardo Hegewisch-Solloa, Janine E Melsen, Hiranmayi Ravichandran, André F. Rendeiro, Aharon G Freud, Bethany Mundy-Bosse, Johannes C Melms, Shira E Eisman, Benjamin Izar, Eli Grunstein, Thomas J Connors, Olivier Elemento, Amir Horowitz, Emily Mace. **Mapping human natural killer cell development in tonsil**. BioRxiv (2023).doi:10.1101/2023.09.05.556371
1. André F. Rendeiro*, Hiranmayi Ravichandran*, Junbum Kim, Alain C. Borczuk, Olivier Elemento, Robert E. Schwartz. **Persistent alveolar type 2 dysfunction and lung structural derangement in post-acute COVID-19**. medRxiv (2022).doi:10.1101/2022.11.28.22282811

Peer reviewed publications

45. Barbara Katharina Geist, Julia Guthrie, Hunor Kertesz, Oana Kulterer, Thomas Nakuz, Lukas Nics, Eva-Maria Patronas, André F. Rendeiro, Chrysoula Vraka, Josef Yu, Stefan Grünert, Marcus Hacker. **Metabolic organ networks derived from PET scans are a novel biomarker for overall health status and identify white adipose tissue as key organ in homeostasis**. Med (2025).doi:10.1016/j.medj.2025.100881
44. Peter Traxler, Stephan Reichl, Lukas Folkman, Lisa Shaw, Victoria Fife, Amelie Nemc, Djurdja Pasajlic, Anna Kusienicka, Daniele Barreca, Nikolaus Fortelny, André F. Rendeiro, Florian Halbritter, Wolfgang Weninger, Thomas Decker, Matthias Farlik, Christoph Bock. **Integrated time series analysis and high-content CRISPR screening delineates the dynamics of macrophage immune regulation**. Cell Systems (2025).doi:10.1016/j.cels.2025.101346
43. Erika Hissong, Bhavneet Bhinder, Junbum Kim, Kentaro Ohara, Hiranmayi Ravichandran, Majd Al Assaad, Sarah Elsoukkary, Michael Shusterman, Uqba Khan, Kenneth Wha Eng, Rohan Bareja, Jyothi Manohar, Michael Sigouros, André F. Rendeiro, Jose Jessurun, Allyson J. Ocean, Andrea Sboner, Olivier Elemento, Juan Miguel Mosquera, Manish A. Shah. **Integrative transcriptomic and single-cell protein characterization of colorectal carcinoma delineates distinct tumor immune microenvironments associated with overall survival**. Pathology - Research and Practice (2025).doi:10.1016/j.prp.2025.156150
42. Felix Eichin, Valentina C Sladky, Matthäus A. Reiner, Marina Leone, Ernesto Abila, André F. Rendeiro, Ralph Böttcher4, Maik Dahlhoff, Thomas Kolbe, Andreas Villunger. **Sequential PIDD1 auto-processing is essential for ploidy control in the liver and heart**. BioRxiv (2025).doi:10.1101/2025.06.23.660994
41. Yimin Zheng, Zhihang Zheng, André F. Rendeiro[Ω], Edwin Cheung[Ω]. **Marsilea: An intuitive generalized visualization paradigm for complex datasets**. Genome Biology (2025).doi:10.1186/s13059-024-03469-3
40. Andreas Reicher, Jií Reini, Maria Ciobanu, Pavel Rika, Monika Malik, Marton Siklos, Viktoriia Kartysh, Tatjana Tomek, Anna Koren, André F. Rendeiro, Stefan Kubicek. **Pooled multicolor tagging for visualizing subcellular protein dynamics**. Nature Cell Biology (2024).doi:10.1038/s41556-024-01407-w
39. Kentaro Ohara, André F. Rendeiro, Bhavneet Bhinder, Kenneth Wha Eng, Hiranmayi Ravichandran, David Pisapia, Aram Vosoughi, Evan Fernandez, Kyrillus Shohdy, Jyothi Manohar, Shaham Beg, David Wilkes, Brian Robinson, Francesca Khani, Rohan Bareja, Scott Tagawa, Andrea Sboner, Olivier Elemento, Bishoy Morris Faltas, Juan Miguel Mosquera. **The evolution of genomic, transcriptomic, and single-cell protein markers of metastatic upper tract urothelial carcinoma**. Nature Communications (2024).doi:10.1038/s41467-024-46320-w
38. Oleksandr Petrenko, Philipp Königshofer, Ksenia Brusilovskaya, Benedikt S. Hofer, Katharina Bareiner, Benedikt Simbrunner, Frank Jühling, Thomas F. Baumert, Joachim Lupberger, Michael Trauner, Stefan G. Kauschke, Larissa Pfisterer, Eric Simon, André F. Rendeiro, Laura P.M. H. de Rooij, Philipp Schwabl, Thomas Reiberger. **Transcriptomic signatures of progressive and regressive liver fibrosis and portal hypertension**. iScience (2024).doi:10.1016/j.isci.2024.109301
37. David J. Falvo, Adrien Grimont, Paul Zumbo, William B. Fall, Julie L. Yang, Alexa Osterhoudt, Grace Pan, André F. Rendeiro, Yinuo Meng, John E. Wilkinson, Friederike Dündar, Olivier Elemento, Rhonda K. Yantiss, Erika Hissong, Richard Koche, Doron Betel, Rohit Chandwani. **A reversible epigenetic memory of inflammatory injury controls lineage plasticity and tumor initiation in the mouse pancreas**. Developmental Cell (2023).doi:10.1016/j.devcel.2023.11.008
36. Jin-Gyu Cheong, Arjun Ravishankar, Siddhartha Sharma, Christopher N Parkhurst, Simon Grassmann, Claire K Wingert, Paoline Laurent, Sai Ma, Lucinda Paddock, Isabella Miranda, Onur Karakaslar, Djamel Nehar-Belaid, Asa Thibodeau, Michael J Bale, Vinay K Kartha, Jim K Yee, Minh Y Mays, Chenyang Jiang, Andrew W Daman, Alexia Martinez de Paz, Dughan Ahimovic, Victor Ramos, Alexander Lercher, Erik Nielsen, Sergio Alvarez-Mullett, Ling Zheng, Andrew Earl, Alisha Yallowitz, Lexi Robbins, Elyse LaFond, Karissa Weidman, Sabrina Racine-Brzostek, He S Yang, David Price, André F. Rendeiro, Hiranmayi Ravichandran, Junbum Kim, Alain C Borczuk, Charles M Rice, R. Brad Jones, Edward J Schenck, Robert J Kaner, Amy Chadburn, Zhen Zhao, Virginia Pascual, Olivier Elemento, Robert E Schwartz, Jason Buenrostro, Rachel E Niec, Franck J Barrat, Lindsay Lief, Joe Sun, Duygu Ucar, Steven Z Josefowicz. **Epigenetic memory of coronavirus infection in innate immune cells and their progenitors**. Cell (2023).doi:10.1016/j.cell.2023.07.019
35. Evan K Noch, Laura N Palma, Isaiah Yim, Nayah Bullen, Yuqing Qiu, Hiranmayi Ravichandran, Junbum Kim, André F. Rendeiro, Melissa B Davis, Olivier Elemento, David J Pisapia, Kevin Zhai, H Carl LeKaye, Jason A Koutcher, Patrick Y Wen, Keith L Ligon, Lewis C Cantley. **Insulin feedback is a targetable resistance mechanism of PI3K inhibition in glioblastoma**. Neuro-Oncology

(2023).doi:10.1093/neuonc/noad117

34. Samir Rustam, Yang Hu, Seyed Babak Mahjour, André F. Rendeiro, Hiranmayi Ravichandran, Andreacarola Urso, Frank DOvidio, Fernando J. Martinez, Nasser K. Altorki, Bradley Richmond, Vasiliy Polosukhin, Jonathan A. Kropski, Timothy S. Blackwell, Scott H. Randell, Olivier Elemento, Renat Shaykhiev. **A Unique Cellular Organization of Human Distal Airways and Its Disarray in Chronic Obstructive Pulmonary Disease**. American Journal of Respiratory and Critical Care Medicine (2023).doi:10.1164/rccm.202207-1384OC

33. Jiwoon Park, Junbum Kim, Tyler Lewy, Charles M. Rice, Olivier Elemento, André F. Rendeiro, Christopher E. Mason. **Spatial omics technologies at multimodal and single cell/subcellular level**. Genome Biology (2022).doi:10.1186/s13059-022-02824-6

32. Junbum Kim, Samir Rustam, Juan Miguel Mosquera, Scott H. Randell, Renat Shaykhiev, André F. Rendeiro$^{\Omega}$, Olivier Elemento$^{\Omega}$. **Unsupervised discovery of tissue architecture in multiplexed imaging**. Nature Methods (2022).doi:10.1038/s41592-022-01657-2

31. Paôline Laurent*, Chao Yang*, André F. Rendeiro*, Benjamin E. Nilsson-payant, Lucia Carrau, Vasuretha Chandar, Yaron Bram, Benjamin R. Tenoever, Olivier Elemento, Lionel B. Ivashkiv, Robert E. Schwartz, Franck J. Barrat. **Sensing of SARS-CoV-2 by pDCs and their subsequent production of IFN-I contribute to macrophage-induced cytokine storm during COVID-19**. Science Immunology (2022).doi:10.1126/sciimmunol.add4906

30. Lissenya B Argueta, Lauretta A Lacko, Yaron Bram, Takuya Tada, Lucia Carrau, André F. Rendeiro, Tuo Zhang, Skyler Uhl, Brienne C Lubor, Vasuretha Chandar, Cristianel Gil, Wei Zhang, Brittany J Dodson, Jeroen Bastiaans, Malavika Prabhu, Sean Houghton, David Redmond, Christine M Salvatore, Yawei J Yang, Olivier Elemento, Rebecca N Baergen, Benjamin R tenOever, Nathaniel R Landau, Shuibing Chen, Robert E Schwartz, Heidi Stuhlmann. **Inflammatory responses in the placenta upon SARS-CoV-2 infection late in pregnancy**. iScience (2022).doi:10.1016/j.isci.2022.104223

29. Jiwoon Park, Jonathan Foox, Tyler Hether, David C. Danko, Sarah Warren, Youngmi Kim, Jason Reeves, Daniel J. Butler, Christopher Mozsary, Joel Rosiene, Alon Shaiber, Evan E. Afshin, Matthew MacKay, André F. Rendeiro, Yaron Bram, Vasuretha Chandar, Heather Geiger, Arryn Craney, Priya Velu, Ari M. Melnick, Iman Hajirasouliha, Afshin Beheshti, Deanne Taylor, Amanda Saravia-Butler, Urminder Singh, Eve Syrkin Wurtele, Jonathan Schisler, Samantha Fennessey, André Corvelo, Michael C. Zody, Soren Germer, Steven Salvatore, Shawn Levy, Shixiu Wu, Nicholas P. Tatonetti, Sagi Shapira, Mirella Salvatore, Lars F. Westblade, Melissa Cushing, Hanna Rennert, Alison J. Kriegel, Olivier Elemento, Marcin Imielinski, Charles M. Rice, Alain C. Borczuk, Cem Meydan, Robert E. Schwartz, Christopher E. Mason. **System-wide transcriptome damage and tissue identity loss in COVID-19 patients**. Cell Reports Medicine (2022).doi:10.1016/j.xcrm.2022.100522

28. Hussein Alnajar*, Hiranmayi Ravichandran*, André F. Rendeiro, Kentaro Ohara, Wael Al Zoughbi, Jyothi Manohar, Noah Greco, Michael Sigouros, Jesse Fox, Emily Muth, Samuel Angiuoli, Bishoy Faltas, Michael Shusterman, Cora N. Sternberg, Olivier Elemento, Juan Miguel Mosquera. **Tumor-Immune Microenvironment Revealed by Imaging Mass Cytometry in a Metastatic Sarcomatoid Urothelial Carcinoma with a Prolonged Response to Pembrolizumab**. Cold Spring Harbor Molecular Case Studies (2022).doi:10.1101/mcs.a006151

27. André F. Rendeiro, Charles Kyriakos Vorkas, Jan Krumsiek, Harjot Singh, Shashi N Kapadia, Luca Vincenzo Cappelli, Maria Teresa Cacciapuoti, Giorgio Inghirami, Olivier Elemento, Mirella Salvatore. **Metabolic and immune markers for precise monitoring of COVID-19 severity and treatment**. Frontiers in Immunology (2021).doi:10.3389/fimmu.2021.809937

26. Nathan C. Sheffield, Micha Stolarczyk, Vincent P. Reuter, André F. Rendeiro. **Linking big biomedical datasets to modular analysis with Portable Encapsulated Projects**. GigaScience (2021).doi:10.1093/gigascience/giab077

25. Laurienne Edgar, Naveed Akbar, Adam T Braithwaite, Thomas Krausgruber, Héctor Gallart-Ayala, Jade Bailey, Alastair L Corbin, Tariq E Khoyratty, Joshua T Chai, Mohammad Alkhalil, André F. Rendeiro, Klemen Ziberna, Ritu Arya, Thomas J Cahill, Christoph Bock, Jurga Laurencikiene, Mark J Crabtree, Madeleine E Lemieux, Niels P Riksen, Mihai G Netea, Craig E Wheelock, Keith M Channon, Mikael Rydén, Irina A Udalova, Ricardo Carnicer, Robin P Choudhury. **Hyperglycaemia Induces Trained Immunity in Macrophages and Their Precursors and Promotes Atherosclerosis**. Circulation (2021).doi:10.1161/CIRCULATIONAHA.120.046464

24. Paul Datlinger*, André F. Rendeiro*, Thorina Boenke, Thomas Krausgruber, Daniele Barreca, Christoph Bock. **Ultra-high throughput single-cell RNA sequencing by combinatorial fluidic indexing**. Nature Methods (2021).doi:10.1038/s41592-021-01153-z
23. Peter Peneder, Adrian Stütz, Didier Surdez, Manuela Krumbholz, Sabine Semper, Mathieu Chicard, Nathan Sheffield, Gaelle Pierron, Eve Lapouble, Marcus Tötzl, Bekir Ergüner, Daniele Barreca, André F. Rendeiro, Abbas Agaimy, Heidrun Boztug, Gernot Engstler, Michael Dworzak, Marie Bernkopf, Sabine Taschner-Mandl, Inge Ambros, Ola Myklebost, Perrine Marec-Berard, Susan Burchill, Bernadette Brennan, Sandra Strauss, Jeremy Whelan, Gudrun Schleiermacher, Christiane Schaefer, Uta Dirksen, Caroline Hutter, Kjetil Boye, Peter Ambros, Olivier Delattre, Markus Metzler, Christoph Bock, Eleni Tomazou. **Multimodal analysis of cell-free DNA whole genome sequencing for pediatric cancers with low mutational burden**. Nature Communications (2021).doi:10.1038/s41467-021-23445-w
22. Johannes C. Melms*, Jana Biermann*, Huachao Huang*, Yiping Wang*, Ajay Nair*, Somnath Tagore*, Igor Katsyv*, André F. Rendeiro*, Amit Dipak Amin*, Denis Schapiro, Chris J. Frangieh, Adrienne M. Luoma, Aveline Filliol, Yinshan Fang, Hiranmayi Ravichandran, Mariano G. Clausi, George A. Alba, Meri Rogava, Sean W. Chen, Patricia Ho, Daniel T. Montoro, Adam E. Kornberg, Arnold S. Han, Mathieu F. Bakhoum, Niroshana Anandasabapathy, Mayte Suárez-Fariñas, Samuel F. Bakhoum, Yaron Bram, Alain Borczuk, Xinzheng V.Guo, Jay H. Lefkowitch, Charles Marboe, Stephen. M. Lagana, Armando Del Portillo, Emmanuel Zorn, Glen S. Markowitz, Robert F. Schwabe, Robert E. Schwartz, Olivier Elemento, Anjali Saqi, Hanina Hibshoosh, Jianwen Que, Benjamin Izar. **A molecular single-cell lung atlas of lethal COVID-19**. Nature (2021).doi:10.1038/s41586-021-03569-1
21. André F. Rendeiro*, Hiranmayi Ravichandran*, Yaron Bram, Vasuretha Chandar, Junbum Kim, Cem Meydan, Jiwoon Park, Jonathan Foox, Tyler Hether, Sarah Warren, Youngmi Kim, Jason Reeves, Steven Salvatore, Christopher E. Mason, Eric C. Swanson, Alain C. Borczuk, Olivier Elemento, Robert E. Schwartz. **The spatial landscape of lung pathology during COVID-19 progression**. Nature (2021).doi:10.1038/s41586-021-03475-6
20. Sandra Schick, Sarah Grosche, Katharina Eva Kohl, Danica Drpic, Martin G. Jaeger, Nara C. Marella, Hana Imrichova, Jung-Ming G. Lin, Gerald Hofstätter, Michael Schuster, André F. Rendeiro, Anna Koren, Mark Petronczki, Christoph Bock, André C. Müller, Georg E. Winter, Stefan Kubicek. **Acute BAF perturbation causes immediate changes in chromatin accessibility**. Nature Genetics (2021).doi:10.1038/s41588-021-00777-3
19. André F. Rendeiro, Joseph Casano, Charles Kyriakos Vorkas, Harjot Singh, Ayana Morales, Robert A DeSimone, Grant B Ellsworth, Rosemary Soave, Shashi N Kapadia, Kohta Saito, Christopher D Brown, JingMei Hsu, Christopher Kyriakides, Steven Chui, Luca Cappelli, Maria Teresa Cacciapuoti, Wayne Tam, Lorenzo Galluzzi, Paul D Simonson, Olivier Elemento, Mirella Salvatore, Giorgio Inghirami. **Profiling of immune dysfunction in COVID-19 patients allows early prediction of disease progression**. Life Science Alliance (2020).doi:10.26508/lsa.202000955
18. Alexander Swoboda, Robert Soukup, Oliver Eckel, Katharina Kinslechner, Bettina Wingelhofer, David Schörghofer, Christina Sternberg, Ha T T Pham, Maria Vallianou, Jaqueline Horvath, Dagmar Stoiber, Lukas Kenner, Lionel Larue, Valeria Poli, Friedrich Beermann, Takashi Yokota, Stefan Kubicek, Thomas Krausgruber, André F. Rendeiro, Christoph Bock, Rainer Zenz, Boris Kovacic, Fritz Aberger, Markus Hengstschläger, Peter Petzelbauer, Mario Mikula, Richard Moriggl. **STAT3 promotes melanoma metastasis by CEBP-induced repression of the MITF pathway**. Oncogene (2020).doi:10.1038/s41388-020-01584-6
17. Thomas Krausgruber, Nikolaus Fortelny, Victoria Fife-Gernedl, Martin Senekowitsch, Linda C. Schuster, Alexander Lercher, Amelie Nemc, Christian Schmidl, André F. Rendeiro, Andreas Bergthaler, Christoph Bock. **Structural cells are key regulators of organ-specific immune responses**. Nature (2020).doi:10.1038/s4 020-2424-4
16. Rainer Hubmann, Susanne Schnabl, Mohammad Araghi, Christian Schmidl, André F. Rendeiro, Martin Hilgarth, Dita Demirtas, Farghaly Ali,Philipp B. Staber, Peter Valent, Christoph Zielinski, Ulrich Jäger, Medhat Shehata. **Targeting Nuclear NOTCH2 by Gliotoxin Recovers a Tumor-Suppressor NOTCH3 Activity in CLL**. Cells (2020).doi:10.3390/cells9061484
15. Elizabeth C Rosser, Christopher J.M. Piper, Diana E Matei ,Paul A. Blair, André F. Rendeiro, Michael Orford, Dagmar G. Alber, Thomas Krausgruber, Diego Catalan, Nigel Klein, Jessica J. Manson, Ignat Drozdov, Christoph Bock, Lucy R Wedderburn, Simon Eaton, Claudia Mauri. **Microbiota-Derived**

Metabolites Suppress Arthritis by Amplifying Aryl-Hydrocarbon Receptor Activation in Regulatory B Cells. Cell Metabolism (2020).doi:10.1016/j.cmet.2020.03.003

14. André F. Rendeiro*, Thomas Krausgruber*, Nikolaus Fortelny, Fangwen Zhao, Thomas Penz, Matthias Farlik, Linda C. Schuster, Amelie Nemc, Szabolcs Tasnády, Marienn Réti, Zoltán Mátrai, Donat Alpar, Csaba Bödör, Christian Schmidl, Christoph Bock. **Chromatin mapping and single-cell immune profiling define the temporal dynamics of ibrutinib drug response in CLL**. Nature Communications (2020).doi:10.1038/s41467-019-14081-6
13. Michael Delacher, Charles D Imbusch, Agnes Hotz-Wagenblatt, Jan-Philipp Mallm, Katharina Bauer, Malte Simon, Dania Riegel, André F. Rendeiro, Sebastian Bittner, Lieke Sanderink, Asmita Pant, Lisa Schmidleithner, Kathrin L Braband, Bernd Echtenachter, Alexander Fischer, Valentina Giunchiglia, Petra Hoffmann, Matthias Edinger, Christoph Bock, Michael Rehli, Benedikt Brors, Christian Schmidl, Markus Feuerer. **Precursors for Nonlymphoid-Tissue Treg Cells Reside in Secondary Lymphoid Organs and Are Programmed by the Transcription Factor BATF**. Immunity (2020).doi:10.1016/j.immuni.2019.12.002
12. Christopher JM Piper, Elizabeth C Rosser, Kristine Oleinika, Kiran Nistala, Thomas Krausgruber, André F. Rendeiro, Aggelos Banos, Ignat Drozdov, Matteo Villa, Scott Thomson, Georgina Xanthou, Christoph Bock, Brigitta Stockinger, Claudia Mauri. **Aryl Hydrocarbon Receptor Contributes to the Transcriptional Program of IL-10-Producing Regulatory B Cells**. Cell Reports (2019).doi:10.1016/j.celrep.2019.10.018
11. Florian Puhm, Taras Afonyushkin, Ulrike Resch, Georg Obermayer, Manfred Rohde, Thomas Penz, Michael Schuster, Gabriel Wagner, André F. Rendeiro, Imene Melki, Christoph Kaun, Johann Wojta, Christoph Bock, Bernd Jilma, Nigel Mackman, Eric Boilard, Christoph J Binder. **Mitochondria are a subset of extracellular vesicles released by activated monocytes and induce type I IFN and TNF responses in endothelial cells**. Circulation Research (2019).doi:10.1161/CIRCRESAHA.118.314601
10. Sandra Schick, André F. Rendeiro, Kathrin Runggatscher, Anna Ringler, Bernd Boidol, Melanie Hinkel, Peter Májek, Loan Vulliard, Thomas Penz, Katja Parapatics, Christian Schmidl, Jörg Menche, Guido Boehmelt, Mark Petronczki, André C. Müller, Christoph Bock, Stefan Kubicek. **Systematic characterization of BAF mutations provides insights into intracomplex synthetic lethalities in human cancers**. Nature Genetics (2019).doi:10.1038/s41588-019-0477-9
9. Sara Sdelci, André F. Rendeiro, Philipp Rathert, Wanhui You, Jung-Ming G. Lin, Anna Ringler, Gerald Hofstätter, Herwig P. Moll, Bettina Gürtl, Matthias Farlik, Sandra Schick, Freya Klepsch, Matthew Oldach, Pisanu Buphamalai, Fiorella Schischlik, Peter Májek, Katja Parapatics, Christian Schmidl, Michael Schuster, Thomas Penz, Dennis L. Buckley, Otto Hudecz, Richard Imre, Shuang-Yan Wang, Hans Michael Maric, Robert Kralovics, Keiryn L. Bennett, Andre C. Müller, Karl Mechtler, Jörg Menche, James E. Bradner, Georg E. Winter, Kristaps Klavins, Emilio Casanova, Christoph Bock, Johannes Zuber, Stefan Kubicek. **MTHFD1 interaction with BRD4 links folate metabolism to transcriptional regulation**. Nature Genetics (2019).doi:10.1038/s41588-019-0413-z
8. Christian Schmidl*, Gregory I Vladimer*, André F. Rendeiro*, Susanne Schnabl*, Thomas Krausgruber, Christina Taubert, Nikolaus Krall, Tea Pemovska, Mohammad Araghi, Berend Snijder, Rainer Hubmann, Anna Ringler, Kathrin Runggatscher, Dita Demirtas, Oscar Lopez de la Fuente, Martin Hilgarth, Cathrin Skrabs, Edit Porpaczy, Michaela Gruber, Gregor Hoermann, Stefan Kubicek, Philipp B Staber, Medhat Shehata, Giulio Superti-Furga, Ulrich Jäger, Christoph Bock. **Combined chemosensitivity and chromatin profiling prioritizes drug combinations in CLL**. Nature Chemical Biology (2019).doi:10.1038/s41589-018-0205-2
7. Tahsin Stefan Barakat, Florian Halbritter, Man Zhang, André F. Rendeiro, Christoph Bock, Ian Chambers. **Functional dissection of the enhancer repertoire in human embryonic stem cells**. Cell Stem Cell (2018).doi:10.1016/j.stem.2018.06.014
6. Paul Datlinger, André F. Rendeiro*, Christian Schmidl*, Thomas Krausgruber, Peter Traxler, Johanna Klughammer, Linda C Schuster, Amelie Kuchler, Donat Alpar, Christoph Bock. **Pooled CRISPR screening with single-cell transcriptome readout**. Nature Methods (2017).doi:10.1038/nmeth.4177
5. Roman A Romanov, Amit Zeisel, Joanne Bakker, Fatima Girach, Arash Hellysaz, Raju Tomer, Alán Alpár, Jan Mulder, Frédéric Clotman, Erik Keimpema, Brian Hsueh, Ailey K Crow, Henrik Martens, Christian Schwindling, Daniela Calvigioni, Jaideep S Bains, Zoltán Máté, Gábor Szabó, Yuchio Yanagawa, Ming-Dong Zhang, André F. Rendeiro, Matthias Farlik, Mathias Uhlén, Peer Wulff, Christoph Bock, Christian Broberger, Karl Deisseroth, Tomas Hökfelt, Sten Linnarsson, Tamas L Horvath, Tibor Harkany. **Molecular**

interrogation of hypothalamic organization reveals distinct dopamine neuronal subtypes. Nature Neuroscience (2016).doi:10.1038/nn.4462

4. Clara Jana-Lui Busch, Tim Hendrikx, David Weismann, Sven Jäckel, Sofie M. A. Walenbergh, André F. Rendeiro, Juliane WeiSSer, Florian Puhm, Anastasiya Hladik, Laura Göderle, Nikolina Papac-Milicevic, Gerald Haas, Vincent Millischer, Saravanan Subramaniam, Sylvia Knapp, Keiryn L. Bennett, Christoph Bock, Christoph Reinhardt, Ronit Shiri-Sverdlov, Christoph J. Binder. **Malondialdehyde epitopes are sterile mediators of hepatic inflammation in hypercholesterolemic mice**. Hepatology (2017).doi:10.1002/hep.28970
3. André F. Rendeiro*, Christian Schmidl*, Jonathan C. Strefford*, Renata Walewska, Zadie Davis, Matthias Farlik, David Oscier, Christoph Bock. **Chromatin accessibility maps of chronic lymphocytic leukaemia identify subtype-specific epigenome signatures and transcription regulatory networks**. Nature Communications. 7:11938 (2016).doi:10.1038/ncomms11938
2. Christian Schmidl*, André F. Rendeiro*, Nathan C Sheffield, Christoph Bock. **ChIPmentation: fast, robust, low-input ChIP-seq for histones and transcription factors**. Nature Methods (2015).doi:10.1038/nmeth.
1. Michaela Schwaiger, Anna Schönauer, André F. Rendeiro, Carina Pribitzer, Alexandra Schauer, Anna Gilles, Johannes Schinko, David Fredman, and Ulrich Technau. **Evolutionary conservation of the eumetazoan gene regulatory landscape**. Genome Research (2014).doi:10.1101/gr.162529.113

Additional publications

2. Dörte Symmank, Felix Clemens Richter, André F. Rendeiro. **Navigating the thymic landscape through development: from cellular atlas to tissue cartography**. Genes & Immunity (2024).doi:10.1038/s41435-024-00257-8
1. Fabian J. Theis, Daniel Dar, Roser Vento-Tormo, Sanja Vickovi, Linghua Wang, Luciane T. Kagohara, André F. Rendeiro, Johanna A. Joyce. **What do you most hope spatial molecular profiling will help us understand? Part 1**. Cell Systems (2023).doi:10.1016/j.cels.2023.05.009

Patents

Approval pending

1. **Methods for determining biological age and/or biological age gap**. *European Patent Office*, 2024-10. Relates to histological aging signatures for disease prediction

Communications

Conference talks

28. **Deep learning for histological image analysis and data privacy**. *Trustworthy AI Summit*, 2026-03. Vienna, Austria
27. **Spatial biology of human aging**. *Center for Pathobiochemistry and Genetics, Medical University of Vienna*, 2026-03. Vienna, Austria
26. **Tissue-specific aging signatures as predictors of disease vulnerability**. *Boehringer Ingelheim, Germany*, 2025-12. Online
25. **Biological image analysis across anatomical scales**. *Machine-Learning Assisted Image Reconstruction workshop of the Austrian Academy of Sciences*, 2025-11. Vienna, Austria
24. **Leveraging spatial biology insights across anatomical scales**. *2nd European CyTOF conference*, 2025-11. Vienna, Austria
23. **Leveraging cross anatomical aging biology for target discovery**. *Boehringer Ingelheim, Austria*, 2025-10. Vienna, Austria
22. **Panel discussion on "PHOTOGRAPHY, MEDICINE AND POWER" with Fiona Tan (Artist) and Monika Pietrzak-Franger (Scientist).**. *Foto.Wien, Arsenal*, 2025-10. Vienna, Austria
21. **Histological aging signatures enable tissue-specific disease prediction from blood**. *17th OeGMBT annual meeting*, 2025-09. Innsbruck, Austria
20. **Histological aging signatures enable tissue-specific disease prediction from blood**. *Department of Pathology*, 2025-01. Innsbruck, Austria
19. **The histological basis of biological aging**. *The annual conference of the Austrian Platform for Precision*

Medicine, 2024-12. Vienna, Austria

18. **A cross anatomical view of aging biology and pathology**. *The IX Siena Think Tank. A vision of Immunooncology*, 2024-10. Siena, Italy
17. **A cross anatomical view of aging biology and pathology**. *VitaDAO minisymposium: Aging across scales and systems*, 2024-06. Vienna, Austria
16. **A cross anatomical view of aging biology and pathology**. *Department of Mathematics, University of Oxford*, 2024-05. Oxford, UK
15. **AI and histopathology: State-of-the-art and next horizons**. *2024 EU-LIFE community meeting*, 2024-04. Vienna, Austria
14. **Tissue-level Prediction of Biological Age and Pathology Incidence**. *Keystone Symposia on Single-Cell Biology: Tissue Genomics, Technologies and Disease*, 2024-01. Whistler, Canada
13. **Spatially and temporally resolved COVID-19 pathology**. *33rd Annual Conference of the German Society for Cytometry*, 2023-10. Berlin, Germany
12. **Multiplexed and spatial biology: connecting spatial scales**. *3rd International Danube Symposium on Whole Person Research*, 2023-09. Vienna, Austria
11. **Unsupervised discovery of tissue architecture with graphs**. *Biological Data Science Meeting, Cold Spring Harbour Laboratory*, 2022-10. Cold Spring Harbor, USA
10. **Spatial Analysis of Tissues and Organs**. *VBC PhD Symposium Pushing Boundaries*, 2022-10. Vienna, Austria
9. **The spatial landscape of lung pathology during COVID-19 progression**. *IMC Summit*, 2021-10. Online
8. **Chromatin mapping and single-cell immune profiling define the temporal dynamics of Ibrutinib response in CLL**. *Young Scientist Association of the Medical University of Vienna PhD Symposia*, 2019-06. Vienna, Austria
7. **Chromatin mapping and single-cell immune profiling define the temporal dynamics of Ibrutinib response in CLL**. *Frontiers in Single Cell Genomics Meeting - Cold Spring Harbour Asia*, 2018-11. Suzhou, China
6. **CROP-seq: updates on the single cell CRISPR screening method**. *10X User Group Meeting 2018*, 2018-04. Heidelberg, Germany
5. **Pooled CRISPR screening with single-cell transcriptome readout**. *SLAS 2018*, 2018-02. San Diego, USA
4. **Pooled CRISPR screening with single-cell transcriptome readout**. *Illumina User Group Meeting 2017*, 2018-02. Online
3. **Large-scale ATAC-seq profiling to identify disease subtypes, regulatory networks and monitoring treatment in CLL**. *Illumina User Group Meeting 2017*, 2018-02. Cologne, Germany
2. **Pooled CRISPR screening with single-cell transcriptome readout**. *Ascona Workshop 2017*, 2017-05. Ascona, Switzerland
1. **Evolutionary conservation of the eumetazoan gene regulatory landscape**. *XVIII Portuguese Genetics Society Meeting*, 2013-06. Porto, Portugal

Conference posters

7. **A cross anatomical scale view of human tissue biology and aging**. *European Society for Spatial Biology (ESSB) meeting*, 2025-10. Heidelberg, Germany. doi:10.6084/m9.figshare.30363898
6. **A cross anatomical scale view of human tissue biology and aging**. *AITHYRA Symposium: AI for Life Science*, 2025-09. Vienna, Austria. doi:10.6084/m9.figshare.30363898
5. **Chromatin mapping and single-cell immune profiling define the temporal dynamics of ibrutinib drug response in chronic lymphocytic leukemia**. *SCOG Workshop Computational Single Cell Genomics*, 2019-05. Munich, Germany. doi:10.6084/m9.figshare.7892663.v1
4. **Combined chromatin accessibility and chemosensitivity profiling identifies targetable pathways and rational drug combinations in Ibrutinib-treated chronic lymphocytic leukemia**. *Young Scientist Association of the Medical University of Vienna PhD Symposia*, 2017-06. Vienna, Austria.
3. **Large-scale chromatin profiling uncovers heterogeneity of molecular phenotypes and gene**

regulatory networks of chronic lymphocytic leukemia. *Young Scientist Association of the Medical University of Vienna PhD Symposia*, 2016-06. Vienna, Austria. doi:10.6084/m9.figshare.3479528.v1 **Best poster award in Malignant Diseases category**

2. **Large-scale chromatin profiling uncovers heterogeneity of molecular phenotypes and gene regulatory networks of chronic lymphocytic leukemia**. *Keystone Symposia on Chromatin and Epigenetics*, 2016-03. Whistler, Vancouver, Canada. doi:10.6084/m9.figshare.3479528.v1
1. **Identification of cis-regulatory elements in the sea anemone Nematostella vectensis**. *Evonet Symposium*, 2012-09. Vienna, Austria. doi:10.6084/m9.figshare.107026

Additional experience

Teaching

2025 - present Invited lecture: Weill Cornell Medicine clinical genomics course: Single-Molecule and Spatial Sequencing: Methods and Applications

2024 - present Bioinformatics course, CeMM PhD Program, Vienna - co-organizor

2022 - present Bioinformatics course, CeMM PhD Program, Vienna - one day workshop

2022 - present Spatial Analysis of Tissues and Organs lecture, CeMM PhD Program, Vienna

2022 - present Invited lecture: Spatial Analysis of Tissues and Organs (Biomedical Informatics and Genomic Medicine, MSc Molecular Precision Medicine, Medical University of Vienna)

Supervision and Mentoring

2026/03 - Jessica Xu, Master student - Boehringer Ingelheim

2025/09 - Shreshta Srivastava, PhD student - LBI-NetMed

2025/08 - Parijat Chakraborty, Postdoc - LBI-NetMed

2023/05 - 2025/11 Sabina Tepus, Master student - Boehringer Ingelheim

2024/09 - Ariadna Villanueva Marijuan, PhD student - CeMM

2024/02 - Lisa Kleissl, Postdoc - CeMM

2023/10 - Yimin Zheng, Postdoc - CeMM

2023/09 - Tamas Veres, PhD student - CeMM

2023/07 - 2023/07 Alessandro Rodia, High school student intern - CeMM

2023/05 - Gabriel Meca Laguna, Master student - SENS Research Foundation

2022/09 - Ernesto Abila, PhD student - CeMM

2022/09 - Iva Buljan, PhD student - CeMM

2022/01 - 2022/04 Zhuoran (Karen) Xu, Data Science Statistician - Weill Cornell Medical College

2021/07 - 2022/10 Kelsey Chetnik, Staff Associate - Weill Cornell Medical College

2020/09 - 2024/03 Junbum (June) Kim, Graduate student - Weill Cornell Medical College

Courses attended

2026 PI leadership course - CeMM - Austria

2023 ERC grant writing masterclass - EU-Life - EU

2022 EMBO Lab Leadership Course - EMBO - Europe

2022 Motivation and Guidance of students during diploma or PhD thesis - Medical University of Vienna - Austria

2022 PI Crash Course: Skills for Future or New Lab Leaders - Columbia University - USA

2021 Probabilistic Modeling in Genomics - Cold Spring Harbor - USA

2015 Summer School on Machine Learning for Personalised Medicine - Marie Curie ITN - UK

2012 Scientific writing course - University of Aveiro - Portugal

Associative/Administrative

2010/09 - 2012/06 Member of the Biology department counsel, University of Aveiro, Portugal

2009/09 - 2011/06 Member of the undergraduate Biology committee, University of Aveiro, Portugal

Software

LazySlide	A package for modular and scalable whole slide image analysis https://github.com/RendeiroLab/LazySlide
WSIData	Efficient data structures and IO for whole slide image analysis https://github.com/RendeiroLab/WSIData
wsi	A Python package for the processing of whole slide histopathological images https://github.com/RendeiroLab/wsi
cytomine_utils	A package with utilities to interact with Cytomine APIs https://github.com/RendeiroLab/cytomine_utils
IMC	A package for the analysis of imaging mass cytometry data https://github.com/ElementoLab/imc
imcpipeline	A pipeline for the preprocessing of imaging mass cytometry data https://github.com/ElementoLab/imcpipeline
imctransfer	Program for the robust, parallel transfer of raw IMC data between machines https://github.com/ElementoLab/imctransfer
page-enrichment	A Python implementation of the Parametric Analysis of Gene Set Enrichment (PAGE) https://github.com/afrendeiro/page-enrichment
ngs-toolkit	A toolkit for the analysis of NGS data https://github.com/afrendeiro/toolkit
peppy	A package to work with Portable Encapsulated Projects (PEP) in Python https://github.com/pepkit/peppy
looper	A job controller for Portable Encapsulated Projects (PEP) https://github.com/pepkit/looper
open_pipelines	Pipelines for a variety of NGS data https://github.com/epigen/open_pipelines
ngstk	A collection of CLI tools for bioinformatics workflows https://github.com/pepkit/ngstk

Licenses and certifications

2021/04 - 2025/04	**Biomedical Research Investigators and Key Personel**, *Credential ID: 41194853* CITI Program
2021/04 - 2025/04	**Good Clinical Practice**, *Credential ID: 41194854* CITI Program
2020/06 - 2026/01	**Responsible Conduct of Research for Faculty** Weill Cornell Medical College
2020/12 - 12-2025	**Responsible Conduct of Research** Tri-Institutional program: MSK Cancer Center, Weill Cornell Medical College, Rockefeller University

Associations

2023 -	Austrian Association of Molecular Life Sciences and Biotechnology (ÖGMBT)
2022 -	Vienna Cell Network
2020 - 2022	New York Academy of Sciences
2015 - 2018	European Hematology Association (EHA)

Updated on 2026-04-17